MILITARY SPECIFICATION

CHARGER, BATTERY, METALLIC RECTIFIER, NAVAL SHIPBOARD, GUIDED MISSILE APPLICATION

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1 SCOPE

1.1 Scope. This specification covers the requirements for the production, inspection and tests for a metallic rectifier type battery charger intended for use in charging the batteries contained in the telemetric data transmitting sets of TERRIER and TARTAR guided missile exercise heads (see 6.1).

2 APPLICABLE DOCUMENTS

2.1 Government-furnished documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

/ FSC 6130 /

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SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901	Shockproof Equipment, Class HI (High Impact) Shipboard Application, Tests for (Navy)
MIL-E-917	Equipment, Electric Power, Basic Requirements for (Naval Shipboard Use)
MIL-A-5092	Adhesive, Rubber (Synthetic and Reclaimed Rubber Base)
MIL-M-10304	Meters, Electrical Indicating, Panel Type, Ruggedized
MIL-G-12803	Gasket Material, Nonmetallic
MIL-S-15291	Switches, Rotary, Snap Action, Bureau Design
MIL-P-15736	Power Supplies, Metallic Rectifier, Naval Shipboard
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery
MIL-C-17588	Circuit Breaker (Automatic-ALB) and Switch, Toggle (Circuit Breaker, Nonautomatic-NLB) Air, Insulated Enclosure, 125 volts and below, AC or DC, Naval Shipboard
MIL-I-26860	Indicator, Humidity, Plug Color Change

DRAWINGS

BUREAU OF NAVAL WEAPONS

1618556	Screwdriver, Torque

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STANDARDS

MILITARY

MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-701	Preferred and Guidance Lists of Semiconductor Devices

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

WR-11	Design and Test of Packaging, Packing, Shipping and Handling Equipment for Weapons Systems Components

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the Naval Supply Depot, Code DCS, 5801 Tabor Ave., Philadelphia 20, Pennsylvania. Copies of drawings and publications should be obtained from Central Technical Documents Office, NOP, Louisville, Kentucky.)

3 REQUIREMENTS

3.1 Description. The battery charger is housed in a metal cabinet with provisions for bolting to a bulkhead, deck or workbench. It is provided with natural ventilation. The battery charger uses 115-volt, 60 cycles per second (c.p.s.) power to develop a regulated, low-voltage d.c. output. The d.c. output voltage is automatically controlled.

3.2 Definitions. Definitions contained in Specification MIL-P-15736 are applicable to this specification.

3.3 General specification. The battery charger shall conform to the applicable requirements of Specification MIL-P-15736 except as specified herein.

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chargers shall not be mechanically

3.4 Performance and product characteristics.

3.4.1 Qualification. The qualification of battery chargers shall not be mandatory.

3.4.2 Input power requirements. The battery charger shall operate with a power source of the following characteristics:

a.	Voltage, a.c.	115 ± 10 per cent
b.	Frequency	60 c.p.s. ± 5 per cent
c.	Phase	Single

3.4.3 Output power requirements. The battery charger shall develop the following output on a continuous duty basis:

a.	Voltage, d.c.	30 to 40
b.	Amperes, d.c.	12, maximum

3.4.3.1 Voltage ripple. The maximum allowable voltage ripple in the output shall not be greater than 5 per cent r.m.s.

3.4.4 Type of load. The battery charger shall be used to charge nickel-cadmium batteries containing from 21 to 26 cells.

3.4.5 Rectifying agent. The rectifying agent shall be silicon rectifiers, selected in conformance with the requirements of Standard MIL-STD-701. Full wave rectification shall be provided.

3.4.6 Operating requirements.

3.4.6.1 Regulation. The d.c. output voltage shall be automatically regulated to within ± 1 per cent regardless of the rated input voltage fluctuations.

3.4.6.2 Voltage selection. The d.c. output voltage shall be adjustable from 30 volts to 40 volts at any load current from no load to the full load of 12 amperes in voltage step increments not greater than 0.5 volt. A fine output voltage adjustment down to 0.1 volt may be provided.

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3.4 Ambient temperature. The rating of the battery

3.4.6.3 Load. The battery charger shall deliver its output to either a complete battery load or to a complete resistive load.

3.4.6.4 Ambient temperature. The rating of the battery charger shall be based upon an ambient temperature range of 23° C. to 50° C.

3.4.6.5 Efficiency. The contractor shall state in his bid or proposal the guaranteed efficiency of the battery charger at no load, half load and full load.

3.4.7 ON-OFF switch. A separate switch shall be provided for operation of the battery charger. The ON-OFF switch shall disconnect both lines of the alternating current (a.c.) input power and at least one line of the direct current (d.c.) output circuit.

3.4.8 Timer. A timer shall be provided which will cut-off the operation of the battery charger 60, plus 5, minus 0 minutes after the initial closing of the ON-OFF switch.

3.4.9 Meters.

3.4.9.1 Voltmeter. The d.c. voltmeter shall be a ruggedized, 3.5-inch round, flush mounted panel type conforming to the requirements of Specification MIL-M-10304. The voltmeter shall have a full range scale of 50 volts and shall be provided with an external zero adjuster.

3.4.9.2 Ammeter. The d.c. ammeter shall be a ruggedized, 3.5-inch round, flush mounted panel type conforming to the requirements of Specification MIL-M-10304. The ammeter shall have a full range scale of 20 amperes and shall be provided with an external zero adjuster.

3.4.10 Circuit breaker. A circuit breaker conforming to the requirements of Specification MIL-C-17588, with a trip element adjustable to suit the load, shall be provided in the input a.c. power lines for the protection of the battery charger.

3.4.11 Switches. Switches shall conform to the requirements of Specification MIL-S-15291 with insulation conforming to the requirements of Specification MIL-E-917

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or as otherwise approved by the procuring activity. Switches shall have a position identification plate to clearly identify the position of operation.

3.4.12 Fuses and fuse holders. A fuse shall be provided in the d.c. output circuit for overload protection. The fuse holder shall be door mounted and shall be of the blown fuse indicator type.

3.4.13 Enclosure. The battery charger shall be completely contained in one enclosure. The enclosure shall be dripproof protected as defined in Standard MIL-STD-108, and shall be portable with provisions for deck, workbench or bulkhead mounting. Material for the enclosure may be steel or aluminum or both in order to meet the weight requirement specified herein. The enclosure shall be provided with a front hinged access door. Circuit breaker, ammeter, voltmeter, ON-OFF switch and fuse holder shall be mounted on the door and shall be recessed such that control knobs and other projections do not extend beyond the front edges of the enclosure as protection against mechanical damage during operation and handling. Louvers shall be provided to ensure adequate natural ventilation.

3.4.14 Size. The battery charger shall conform to the following dimensions:

a.	Height	20 inches maximum
b.	Width	20 inches maximum
c.	Depth	14 inches maximum

3.4.15 Weight. The weight of the battery charger shall not exceed 70 pounds.

3.4.16 Safety. A high degree of personnel safety shall be required. The battery charger output circuit shall be completely insulated from the input circuit. Specifications MIL-P-15736 and MIL-E-917 shall apply.

3.4.17 Shipping condition.

3.4.17.1 Shock transportation, storage and handling. The battery charger after being subjected to the test specified in 4.5.12.1 shall show no damage and shall meet all the requirements of Section 3.

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required by the contract or purchase order.

3.4.17.2 Vibration-transportation. The battery charger after being subjected to the tests specified in 4.5.12.2 shall show no damage and shall meet all the requirements of Section 3.

3.5 Repair parts. On board repair parts shall be furnished in accordance with the requirements of Specification MIL-P-15736. The number of sets or consolidated "shipworth" parts shall be required by the contract or purchase order.

3.6 Drawings. Drawings shall be furnished as required by the contract or purchase order.

3.7 Technical manuals. Technical manuals shall be provided as required by the contract or purchase order.

3.8 Technical data. Technical data shall be provided as required by the contract or purchase order.

3.9 Test procedures. The contractor shall prepare detailed test procedures for all tests required in Section 4 of this specification and shall submit them to the Bureau of Naval Weapons for approval prior to production. These test procedures shall include a list of test equipment required to conduct the tests.

3.10 Workmanship. The battery charger, including all parts and components, shall be fabricated and finished in a manner to assure compliance with all requirements of this specification. The requirements for first class workmanship do not preclude the use of time-and-labor-saving techniques commensurate with the nature of the assembly, provided that reliability is in no way jeopardized.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility. The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or purchase order. The Government reserves the right to perform any of the inspections and tests set forth in the specification where such inspections and tests are deemed necessary to assure that the supplies and services conform to the prescribed requirements.

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4.2 Sampling.

4.2.1 Lot. A lot shall be as defined in Specification MIL-P-15736.

4.2.2 Sampling for inspection, routine and selective tests. Sampling for inspection, routine and selective tests shall be in accordance with Tables III and IV of Specification MIL-P-15736.

4.2.3 Periodic tests. The first equipment of the design offered for delivery under the contract or purchase order shall be subjected to the periodic tests.

4.3 Inspection.

4.3.1 Battery chargers. In addition to the requirements for inspection of power supplies as specified in Specification MIL-P-15736, battery chargers shall be inspected for painting and for workmanship.

4.3.2 Repair parts. Repair parts shall be inspected as specified in Specification MIL-P-15736.

4.4 Tests.

4.4.1 Routine tests. Each sample battery charger selected in accordance with 4.2.2 shall be subjected to the routine tests specified in Table V of Specification MIL-P-15736.

4.4.1.1 Repair parts. Resistance readings at room temperature shall be made upon all repair coils, resistors, rheostats and potentiometers and the values checked against the values obtained for the same part in the battery charger. Coils, resistors, rheostats, potentiometers, rectifier stacks and switches shall be submitted to the dielectric test.

4.4.2 Selective tests. Each sample battery charger selected in accordance with 4.2.2 shall be subjected to the selective tests specified in Table V of Specification MIL-P-15736 except that the starting test shall not be required.

4.4.3 Periodic tests. The battery charger, selected in accordance with 4.2.3, shall be subjected to the periodic tests of Table V of Specification MIL-P-15736 except that the starting and phase balance tests shall not be required.

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4.5 Test details.

4.5.1 Efficiency. The efficiency of the battery charger shall be measured as specified in Specification MIL-P-15736. The efficiency shall equal or exceed the value guaranteed by the contractor (3.4.6.5).

4.5.2 Insulation resistance. When tested as specified in Specification MIL-P-15736, the insulation resistance shall be not less than 10 megohms.

4.5.3 Enclosure. The battery charger shall satisfactorily meet the requirements of the test for dripproof protected enclosure as specified in Standard MIL-STD-108.

4.5.4 Heating. A heating test shall be conducted as specified in Specification MIL-P-15736. The maximum steady state temperature observed shall not be greater than the maximum allowable for the type of insulation used in each application.

4.5.5 Weight. The weight of the completely assembled battery charger shall not exceed 70 pounds.

4.5.6 Inclined operation. The battery charger shall be tested for inclined operation into a resistive load at full load as specified in Specification MIL-P-15736.

4.5.7 Shock. The class HI shock tests (electrical operation and mechanical damage) shall be conducted in accordance with Specification MIL-S-901 for medium weight equipment with test features as specified in Specification MIL-P-15736.

4.5.8 Vibration. Vibration tests shall be conducted as specified in Specification MIL-P-15736.

4.5.9 D.C. output. The battery charger shall be tested for d.c. output as required by Specification MIL-P-15736.

4.5.9.1 Regulation. Regulation shall be tested in accordance with Table VI of Specification MIL-P-15736 and shall not be greater than ± 1 per cent.

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4.5.9.2 Voltage selection. The battery charger shall be tested for selection of output voltages between the values of 30 and 40 volts.

4.5.9.3 Ripple voltage. The ripple voltage, when measured in accordance with Specification MIL-P-15736, shall not be greater than 5 per cent.

4.5.9.4 Timer. When operating into a resistive load at full load, the timer shall operate to cut off the battery charger at the expiration of 60 plus 5 minus 0 minutes.

4.5.10 Creepage and clearance distances. Creepage and clearance distances shall be demonstrated by actual measurement to be in accordance with the requirements of Specification MIL-E-917.

4.5.11 Dielectric tests. Dielectric tests of the battery charger shall be conducted in accordance with Specification MIL-P-15736.

4.5.12 Shipping condition.

4.5.12.1 Shock-transportation, storage and handling. The battery charger packaged and packed as specified in Section 5, shall be subjected to the appropriate shock isolation tests specified in Weapons Requirements WR-11.

4.5.12.2 Vibration-transportation. The battery charger packaged and packed as specified in Section 5, shall be vibration tested as specified in Weapons Requirement WR-11.

4.5.12.3 Functional tests. At the conclusion of the tests specified in 4.5.12.1 and 4.5.12.2 above, functional and other tests, as applicable, shall be conducted on the battery charger to determine freedom from operational malfunction and compliance with all requirements of Section 3.

4.5.12.4 Waiver of tests. Tests specified in 4.5.12.1 and 4.5.12.2 above may be waived by the procuring agency for level C packs only, provided the contractor guarantees safe delivery to the using activity.

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5 PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Not applicable.

5.1.1 Level A.

5.1.1.1 Cleaning, drying and preservative application. Not applicable.

5.1.1.2 Unit packaging Unit packaging of the battery charger shall be in accordance with Method IIa of Specification MIL-E-17555. The unit package shall contain one battery charger. Gasket material shall be in accordance with Specification MIL-G-12803, type 1, cemented in place with adhesive conforming to Specification MIL-A-5092, type III. The humidity indicator shall be in accordance with Specification MIL-I-26860, type II. The rough handling tests of Specification MIL-P-116 shall not apply.

5.1.1.3 Intermediate package. Not applicable.

5.1.1.4 Cushioning. The unit package, and pack, shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices, to ensure that the battery charger will meet all the requirements of Section 3 after being tested in accordance with 4.5.12.1 and 4.5.12.2.

At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5%.

5.1.2 Level B. Not applicable.

5.1.3 Level C.

5.1.3.1 Cleaning, drying and preservative application. Not applicable.

5.1.3.2 Unit packaging. Unit packaging of the battery charger shall be in accordance with Method III of Specification MIL-P-116. The unit package shall contain one battery charger.

5.1.3.3 Intermediate package. Not applicable.

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5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior containers. The battery charger packaged as described in 5.1.1.2 shall require no over-packing.

5.2.2 Level C. The battery charger packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Special markings. None, unless otherwise specified.

5.3.2 Normal markings. In addition to the markings required by contract or order, unit packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6 NOTES

6.1 Intended use. The battery charger is intended to be used for charging the batteries contained in the telemetric data transmitting sets of the Exercise Heads Mk 5 Mods 2 and 3 and Mk 7 Mod 2, which, in turn, are used with TERRIER BT, BTN, HT and Improved TARTAR missiles. These batteries are the nickel-cadmium type containing 22 cells.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. The number of battery chargers to be furnished
c. The number of sets of repair parts or consolidated "shipboard" spares (3.5)
d. The drawings required (3.6)
e. The technical manuals required and the number thereof (3.7)
f. The technical data required (3.8)
g. The levels required in the preparation for delivery (5)
h. Change approval procedures
i. All options or deviations from this specification

6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116 and to par. 4.5.12.4.

6.2.2 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

For Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.3 Requests for deviations and waivers. Each request that a deviation or waiver be granted shall be submitted to the procuring activity in writing. Each request and the reply by which a deviation or waiver is granted by the Bureau of Naval Weapons or by a naval activity which has been delegated such authority shall state specifically the extent of the departure, the number of items affected, the identifying number of the item, the contractor and the contract number.

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6.4 Conflicting requirements. Conflicting requirements arising between this specification and any specifications, publications or drawings listed herein shall be referred in writing to the procuring activity or its appointed agent for interpretation and clarification.

6.5 Sample procedure.

Discharge-Charge Procedure.

1. Visually inspect each battery for physical defects such as cracks, leaks, etc.

   Note: If battery shows evidence of electrolyte leakage, clean battery surfaces with distilled water prior to removal and inspection of vent caps. If vent caps indicate damage to gaskets, replace gaskets before re-issuing battery.

2. Remove bent caps; inspect and clean vent holes. Do not re-install caps until de-gassing period is completed.

3. Connect battery (refer to Figure 1) to a 10 ohm 25 watt resistor and discharge until the 3 volt level is reached.

4. With battery at 3 volts or below, short individual cells with continuous wrapping of #18 bus wire for at least 60 minutes (one hour). Do not unscrew terminal nuts.

5. After discharge of each battery is completed, remove jumper wire and connect the two batteries in series. Connect positive terminal of one battery to negative terminal of other with wire size #8 or equivalent (refer to Figure 2).

   Note: If battery charging equipment contains a voltmeter and an ammeter disregard Step 6 and proceed to Step 7.

6. Connect battery charger, voltmeter, ammeter, and batteries as indicated in Figure 2.

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6.5 Sample procedure (continued)

7. If charging equipment contains a voltmeter and ammeter, attach negative terminal of one battery to negative lead of charger, and positive terminal of other battery to positive lead of charger.

8. Activate charger and allow ten minute warmup period. Apply charging voltage to battery from charger in accordance with operating instructions for a period of 60 $^{+5}_{-0}$ minutes.

   Note: Charger should be adjusted to output voltage of 34.4 volts. Check voltage adjustment prior to full charge cycle. Batteries should be charged at temperature range between + 45° F. and + 95° F. in well-ventilated area.

9. After completion of one-hour's charge, turn off charger and disconnect batteries. Place batteries in upright position in well-ventilated area.

10. Allow battery to de-gas for a period of 12 hours and record de-gassing time.

CAUTION

Battery must remain upright until after 12-hour de-gassing period.

After de-gassing, check electrolyte level of each cell. Level should not be lower than 1/16 inch below fill line. If lower than specified, add distilled water until fill line is reached. No topping-off charge is required after adding distilled water.

11. Replace vent caps removed in Step 2 after determining that vent cap hole is clear. Tighten caps with 7 #1 in/lb of torque using Torque Screwdriver #1618556.

12. Connect one battery pack (Figure 3) to perform load test.

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13. Discharge battery for 15 minutes, measuring voltage of battery under load during this time. Voltage must be 12.0 volts or greater at end of 15-minute period.

    a. If voltage is satisfactory, discharge battery as described in Steps 3 and 4, attach jumpers, and return battery to stock or install in telemeter insert.

    b. If voltage is less than 12 volts, reject battery. Discharge completely as described in Steps 3 and 4, install jumper, and return battery to supply system with a written explanation of rejection.

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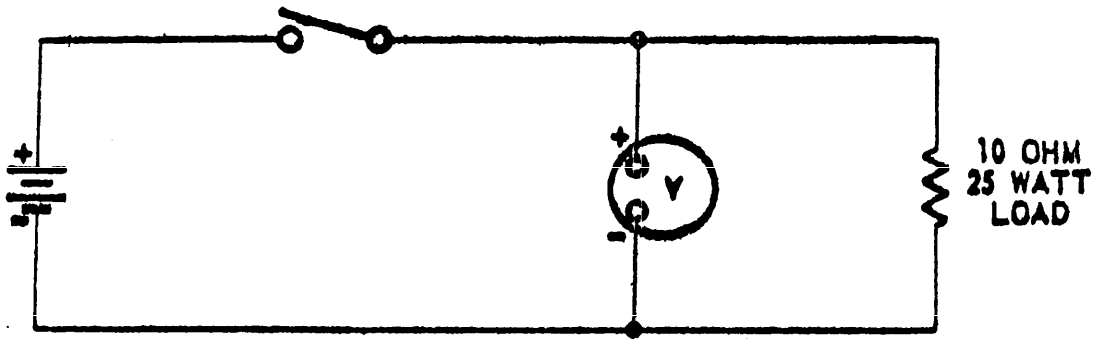


FIGURE 1

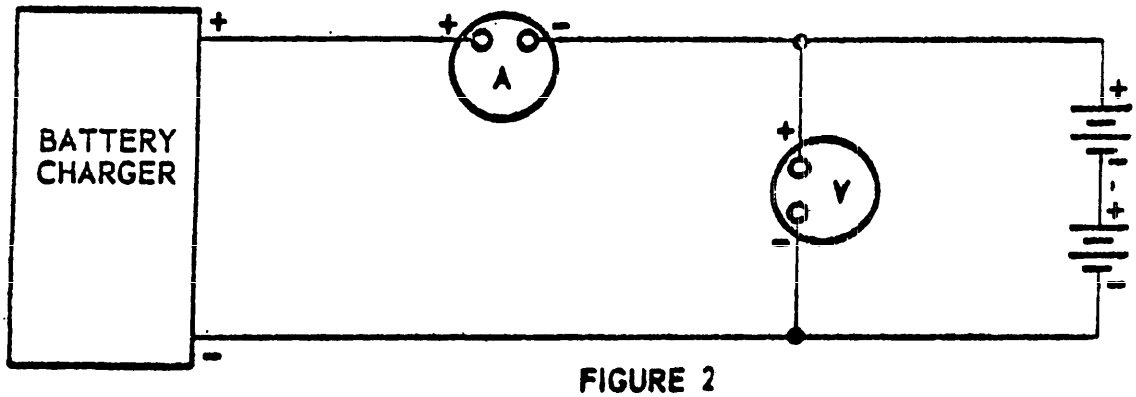


FIGURE 2

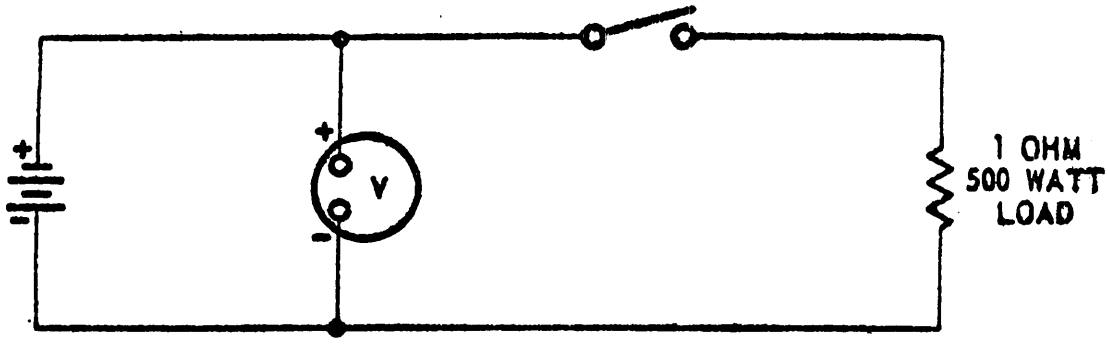


FIGURE 3

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